

Assignment for "Human Motion Computing"

Due: July 20th, 2026

Human Motion Computing studies how human motion can be sensed, modeled, interpreted and supported using AI techniques. In this assignment, you will analyze two complementary approaches for computationally modeling human movement:

- **Inertial sensing**, where movement is represented as multivariate time series captured by wearable sensors.
- **Image-based sensing**, where movement is represented through human pose estimation and skeletal models extracted from images or videos.

Generative AI Policy

- Generative AI Tools **CAN** be used to support the usage of software tools in **Part I and Part II – Approach B**. You should indicate the tools used and for what purpose.
- Generative AI Tools **CANNOT** be used for critically reading the papers provided to do **Part II – Approach A** (nor answering the questions in the other tasks).

You have to choose ONE of the three options: Part I, Part II-A OR Part II-B.

Support: You can contact professor Olga Santos (ocsantos@dia.uned.es) for questions, hints or suggestions about the assignment.

Part I – Human Motion Computing with Inertial Signals

Background

The KSAS dataset [1] contains smartphone inertial sensor data collected during the execution of American Kenpo Karate Blocking Set I. The XROCKET algorithm [2] was used in [3] to classify Aikido practitioners' expertise while providing interpretable explanations of the classification decisions. The same approach was also followed in [4] to differentiate early stages of cognitive decline and advanced dementia based on the movements performed while sleeping.

Your task is to apply XROCKET to the KSAS dataset (or any other dataset you know or create) to extract meaningful information from the motion data collected. Details about the dataset are available in [5]. The implementations done by PhyUM researchers are available in [6] and [7] as support (but have not been exhaustively tested).

Links

- [1] KSAS Dataset: <https://github.com/Physical-User-Modeling-PhyUM/KSAS-Dataset>
- [2] Time series classification with XROCKET: <https://dida.do/blog/explainable-time-series-classification-with-x-rocket>
- [3] UMAP 2026 paper (to be published in UMAP proceedings: [umap26-45.pdf](#))
- [4] Sleep analysis paper: <https://link.springer.com/article/10.1007/s10796-026-10736-0>
- [5] Aikido paper: <https://link.springer.com/article/10.1007/s11257-024-09393-2>
- [6] Code for Sleep analysis: <https://github.com/Physical-User-Modeling-PhyUM/EADS>
- [7] Code for Aikido analysis: https://github.com/Physical-User-Modeling-PhyUM/UMAP26_SP1 (to be available during UMAP conference)

Task 1.1 – Sensor Axes contribution analysis (Where is the information?)

Analyze the XROCKET explanations regarding the information about the IMU axes and respond:

- Which sensor signals contribute most to classify the user movement and/or user performance?

Discussion: Relate your findings to the biomechanics of the movements.

Task 1.2 – Temporal Pattern Analysis: Dilations/Frequencies (At what temporal scale is the information?)

Analyze the dilation values selected by XROCKET. Determine whether the classification relies primarily on:

- short-duration, high-frequency motion patterns,
- long-duration, global movement structures,
- or a combination of both.

Discussion: Explain how temporal scale influences movement recognition.

Task 1.3 – Explainable Human Motion Computing

Interpret the most discriminative patterns identified by XROCKET in the dataset used.

Discussion:

- what aspects of the movement are being captured,
- whether the explanations appear meaningful from a human perspective,
- how these explanations could support learning and performance assessment.

Deliverable

Technical Report (PDF). The report should include:

- Description of the inertial dataset used.
- Description of the steps followed to answer the questions.
- Problems encountered (if any) preparing the implementation.
- Link to the code repository, notebook, etc. produced.
- Responses to each of the tasks, including visualizations computed for the analysis to support the discussion of the findings.
- Use of Generative AI tools to support the software implementation.

Background

Explore three human pose estimation approaches: MediaPipe Pose [8], OpenPose [9] and YOLO Pose [10] and discuss their applicability to the practice of Tai Chi in older adults aimed to foster active ageing [11].

There are 2 options to work this part.

Links

[8] MediaPipe: https://ai.google.dev/edge/mediapipe/solutions/vision/pose_landmarker

[9] OpenPose: <https://openpose.org/>

[10] YOLO Pose: <https://docs.ultralytics.com/tasks/pose>

[11] Active ageing paper: [umap26-35.pdf](#) (to be published in UMAP proceedings)

[12] KLS paper: <https://link.springer.com/article/10.1007/s10055-026-01326-1>

[13] KUMITRON paper: <https://www.mdpi.com/1424-8220/21/24/8378>

[14] SPAF paper: <https://www.mdpi.com/1424-8220/25/11/3436>

[15] OpenCV pose estimation: <https://www.opencv.ai/ai-services/pose-estimation>

APPROACH A:

Your task is to act as the Lead Architect of the PhyUM Research Center, audit the design decisions behind the development of KLS [12], KUMITRON [13] and SPAF [14] and discuss what would be the best approach to build a psychomotor system for older adults to practice Tai Chi. Then present your audit findings to the PhyUM Board.

Task 2.1 – KLS Audit: Symbolic Logic vs. Machine Learning

Analyze the architectural decisions described in the KLS paper.

Question: Why did the authors choose a rule-based approach based on Finite State Machines instead of training a conventional Machine Learning classifier?

Task 2.2 – KUMITRON Audit: The Bottom-Up Necessity

Analyze the Human Pose Estimation strategy adopted in KUMITRON.

Question: Why was OpenPose's bottom-up approach particularly appropriate for karate kumite analysis?

Task 2.3 – SPAF Audit: From Skeletons to Clinical Meaning

Analyze the role of semantic abstraction in SPAF.

Question: Why is pose estimation alone insufficient for clinical assessment? Explain the role of semantic abstraction in transforming skeleton data into clinically meaningful information.

Task 2.4 – Design proposal

Imagine you are designing a new Human Motion Computing system for the Active Ageing domain.

Questions:

- a) Which of the three HPE approaches would you adopt, and why?
- b) Would you prefer to use wearables to collect inertial signals, and why?

Deliverable

Video recording (MP4) of your presentation to the PhyUM Board.

- The video must not exceed 5 minutes.
- Use supporting slides to do the presentation.

APPROACH B:

Develop some code that allows to compare the three HPE approaches discuss in the course (MediaPipe Pose [8], Open Pose [9] and YOLO Pose [10]) as well as the HPE algorithm from by OpenCV [15]. Compare them using different videos of older adults practicing Tai Chi. Which one would you select?

Pay special attention to:

- the keypoints identified by each algorithm and the body regions represented,
- the time required for processing,
- the robustness of the pose estimation under realistic conditions (occlusions, lighting, body orientation, etc.),
- the suitability of each approach for supporting Tai Chi practice in older adults.

Deliverable

Technical Report (PDF). The report should include:

- Description of the video/images dataset used and provide link to it.
- Description of the code implemented and link to use it.
- Analysis of the usage of the four HPE algorithms, compare the results and discuss critically.
- Use of Generative AI tools to support the software implementation.